

Model card - Rent estimation in Berlin

Model Details

- Developed by Edvin Perfundí
- Random Forest Regression model trained to estimate Berlin apartment cold rent (Kaltmiete).

Intended Use

- Intended to be used to get an estimation of rent prices in apartments in Berlin.
- Particularly intended for renters and landlords.
- The model was developed using data representative of the 2025–2026 market period and may become less reliable as rental market conditions change. Predictions for substantially different market periods should therefore be interpreted with caution.

Factors

- Potential relevant factors include area, number of rooms, location, year built, availability, heating, floor, floor number and amenities such as an elevator, a kitchen, a shower, etc.

Metrics

- Evaluation metrics include RMSE, R^2 score, MAE and overfitting gap in terms of R^2 from all sets - training, validation and test.
- Prediction uncertainty is estimated from the distribution of predictions generated by the individual trees in the Random Forest.

Dataset

- The dataset used for this model was collected privately by the 'ImmoWelt' website using a dedicated scraper.
- Personal data like phone numbers and names were not scraped.
- Quasi-identifiers such as full address were scraped, but partial features such as ZIP code and district were used instead.

Training data

- Collected dataset, training data split.

Uncertainty Quantification

- Each prediction is the average of 200 individual tree predictions. The spread of those 200 predictions (standard deviation, or percentile interval) is reported alongside the point estimate as a confidence signal.

Known Limitations

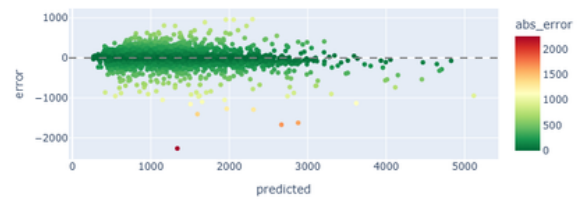
- Systematic underprediction for very expensive apartments (roughly above €3,500/month), where the training data has few examples to learn from.
- District coverage is uneven; districts with very few listings in the training data are likely to produce less reliable estimates.
- Reflects historical ImmoWelt listings at time of training; does not automatically adjust to subsequent market shifts.

Ethical Considerations

- The data gathered and used is publicly available and is used for educational and demonstrative purposes, rather than profit-making purposes.
- This tool is not a substitute for a professional valuation or legal rent assessment, and should not be the sole basis for setting or negotiating an actual rental price.

Quantitative Analysis

Residuals vs predicted value



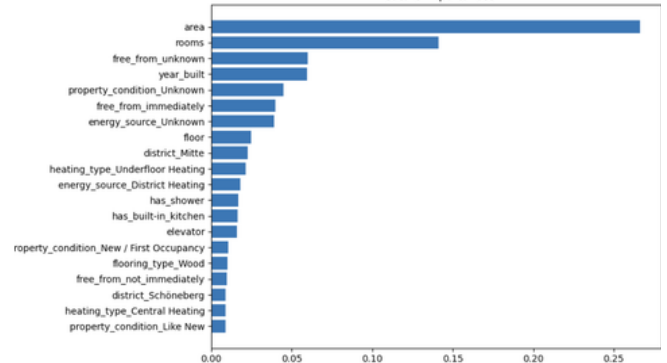
Distribution of residuals



Error by district



Feature Importances



Evaluation data

- Collected dataset, test data split.
- ImmoWelt website new posts for further testing and inference.